

Emotional MMN: Anxiety and heart rate correlate with the ERP signature for auditory change detection

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ABSTRACT

Objective: Previous work established the mismatch negativity (MMN) as a correlate of pre-attentive auditory change detection. The present study aimed at investigating the relationship between the MMN and emotional processes associated with the detection of change.

Methods: To this end, we assessed state anxiety with a questionnaire and subsequently recorded the electroencephalogram (EEG) and heart rate while participants watched a silent movie and listened to a task-irrelevant auditory oddball sequence. The oddball sequence comprised meaningless syllables of which some were deviants spoken with an angry or neutral voice.

Results: The MMN to angry voice deviants was larger than that to neutral deviants and correlated positively with ensuing heart rate acceleration. Additionally, both the MMN and heart rate acceleration to angry voice deviants were increased with increasing state anxiety. A similar effect for neutral voice deviants was non-significant.

Conclusion: Taken together, these results suggest that the pre-attentive processing of threat, as reflected by the MMN, is linked to an activation of the sympathetic nervous system. Moreover, this link is more strongly activated in individuals with high state anxiety.

Significance: Thus, the MMN may be used as a marker for an individual's state dependent sensitivity to unattended, emotionally relevant change.

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1. Introduction

Our environment comprises a multitude of objects that excite the sensory system. We successfully navigate this environment by searching for and attending to a subset of these objects that are important for our current goals. Occasionally, however, an object in the environment is neither searched for nor attended, yet its presence is of consequence to the individual. Researchers have investigated the conditions that determine whether and how individuals respond to such objects. Within the auditory modality, this work has utilized the mismatch negativity (MMN), a neural correlate of pre-attentive change detection (Näätänen et al., 2005, 2007).

The MMN can be observed in the event-related potential (ERP) elicited to rare, unattended sounds and thus is typically studied using a passive auditory oddball paradigm. In such a paradigm, participants are presented with standard sounds that are rarely and unpredictably interrupted by a sound deviating from stan-

dards in one or more physical properties or by a sound that violates an established rule (Korzyukov et al., 2003; Carral et al., 2005). The MMN is revealed by subtracting the ERP of standards from that of deviants. This subtraction isolates a fronto-centrally distributed negativity that peaks between 120 and 200 ms following stimulus onset and whose generators are presumably located in the temporal and frontal lobes (Tse and Penney, 2008). Several studies have found a positive correlation between the MMN and the magnitude of acoustic change (Tiitinen et al., 1994; Novitski et al., 2004). However, further investigations revealed that this relationship is determined by the listener's perception of change magnitude rather than the actual difference between standard and deviant. The larger the perceived change, the greater is the likelihood that an MMN is elicited and the larger is the MMN amplitude in the averaged ERP (Winkler et al., 1995; Horvath et al., 2008; Amenedo and Escera, 2000). As deviants can elicit an MMN while participants are asleep (Ruby et al., 2008) or perform an unrelated task (Escera et al., 2002; Carral et al., 2005), researchers proposed that the MMN reflects pre-attentive auditory change detection (Escera et al., 2002; Carral et al., 2005; Näätänen et al., 2005, 2007; Snyder and Alain, 2007; Ruby et al., 2008). Moreover, the MMN has been attributed to a change detection mechanism by which traces in sensory memory can be updated without conscious awareness

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and by which listeners may be alerted to violations of perceptual expectations derived from past sensory input (Korzyukov et al., 2003; Carral et al., 2005; van Zuijen et al., 2006).

It is easy to conceive the adaptive value of a mechanism that detects unattended changes in the environment. However, not every change is of significance and requires a response from the individual. Thus, one may ask whether pre-attentive change detection is sensitive to this and triggered more easily in case of significant change. In an effort to address this question, researchers studied the elicitation of the MMN to emotional stimuli (De Baene et al., 2004; Goydke et al., 2004; Schirmer et al., 2005, 2007). For example, De Baene and colleagues (2004) presented deviants that differed from standards in their perceived roughness, an acoustic cue that makes sounds inherently unpleasant. Greater differences in perceived roughness between standards and deviants were associated with larger MMN amplitudes. Moreover, perceived roughness was a better predictor for the MMN than its actual acoustic correlate, intensity fluctuation. Thus, the authors concluded that roughness is encoded and perceived pre-attentively. Given that the perception of roughness in a speaker's voice can signal vocal threat (Chasaide and Gobl, 2002), it is not surprising that subsequent work found larger MMN amplitudes to angry as compared to neutral voice deviants (Schirmer et al., 2005, 2007). That a similar MMN difference showed for happy and neutral vocalizations furthermore suggests that, besides roughness, other acoustic emotion cues are encoded pre-attentively (Schirmer et al., 2005, 2007).

Other lines of inquiry that used the MMN to examine emotional processing have focused on autonomic responses or anxiety. Research on autonomic responses demonstrated that neutral, task-irrelevant auditory oddballs can elicit changes in heart rate (HR) or skin conductance (Lyytinen et al., 1992). However, because these changes failed to correlate with the MMN, they were thought to require conscious processing of auditory oddballs (Lyytinen et al., 1992). Moreover, the MMN was proposed to precede rather than to reflect emotional processes. Research on anxiety compared the processing of neutral deviants during safe periods and during periods in which participants anticipated an electric shock (Cornwell et al., 2007). The MMNm, the magnetic counterpart of the MMN, was found to be larger during threat as compared to safe periods suggesting that anxiety sensitizes change detection processes reflected by the MMN. However, research on the effect of anxiolytics on the MMN has failed to consistently support this conclusion (Kasai et al., 2002; but see Rosburg et al., 2004). Moreover, there is evidence that socially threatening situations such as giving a public speech reduce rather than increase the MMN (Simoes et al., 2007).

In sum, past research has established the MMN as a marker of pre-attentive change detection. However, whether and how such change detection interacts with emotional processes is still unclear. While studies comparing emotional and neutral auditory oddballs found that emotional significance modulates MMN amplitude, converging evidence from autonomic and anxiety research is still lacking. One potential explanation for this inconsistency is that the latter type of studies used oddball sequences with small or simple physical changes (e.g., pitch increase). Such changes are of little significance for the listener and may thus fail to reliably excite autonomic reactions and to be amplified by anxiety. Moreover, in order to discover a relationship between bodily arousal, anxiety and the MMN, large or complex changes that have intrinsic emotional significance may be necessary.

With the present study, we further probed the relationship between the MMN and emotional processes. In particular, we were interested in whether the MMN amplitude correlates with natural variations in anxiety as well as sympathetic activation assessed through HR. To this end, we presented participants with a se-

quence of meaningless syllables. In one block, standard syllables were spoken with a neutral tone of voice and deviants with an angry tone of voice. In another block, tone of voice for standards and deviants was reversed. Based on prior work (De Baene et al., 2004; Schirmer et al., 2005, 2007), we predicted that the MMN would be larger to deviants spoken with an angry as compared to a neutral tone of voice. Furthermore, we expected the MMN to correlate with both anxiety and HR measures for angry voice deviants but not or to a lesser degree for neutral voice deviants.

2. Methods

2.1. Participants

Twenty-three participants gave informed consent prior to participation. As there are sex differences in the experience of anxiety (e.g., Abdel-Khalek and Maltby, 2009; Spielberger et al., 1983; Vesga-López et al., 2008) that could potentially confound the relationship between anxiety and the MMN, we selected only female participants. The data from three participants were discarded due to technical defects or EEG artefacts. Of the remaining participants, two were left handed and 18 were right handed according to the Edinburgh Handedness Inventory (Oldfield, 1971). The participants ranged in age from 19 to 21 years. They reported normal or corrected to normal vision, normal hearing and no psychological impairments.

2.2. Material

The stimulus material consisted of the syllables “dada” spoken with neutral or angry prosody. They were taken from a previous study in which they were validated for their emotional character (Schirmer et al., 2005). Both, the neutral and the angry voice tokens were 557 ms long and had a comparable maximum (67 dB) and mean intensity (56 dB; Boersma, 2001). They were presented in two blocks with 1050 standards ($p = 0.875$) and 150 deviants ($p = 0.125$) in each block. There was a minimum of three and a maximum of 11 standards between deviants. The neutral and the angrily spoken syllables served as standards and deviants in two separate blocks. In one block, neutral syllables were presented as standards and angry syllables as deviants. In the other block, angry syllables were presented as standards and neutral syllables as deviants. This intermixed design allowed us to record standard and deviant processing of angrily and neutrally spoken syllables such that each syllable could serve as its own acoustic control (Jacobsen and Schröger, 2003). While past HR investigations typically used a stimulus onset asynchrony (SOA) of several seconds as to allow HR to return to baseline, the SOA in the present study was only 1200 ms long as SOAs of several seconds are inappropriate for the kind of paradigm applied here (Lyytinen et al., 1992). However, as no HR changes were expected to standard stimuli, the critical interval was that between deviants. Moreover, in line with previous work (Lyytinen et al., 1992), deviant onsets were separated by a minimum of 4.8 and a maximum of 14.4 s.

2.3. Procedure

Rather than manipulating anxiety within participants by presenting a threat and a no-threat condition, the present study relied on natural variations in state anxiety induced by a psychological testing situation (Spielberger et al., 1983). This allowed us to reduce experimental time, which was already 48 min because of the two standard and deviant conditions used here. Thus, before commencing the experiment, participants were asked to complete the state anxiety scale of the State-Trait Anxiety Inventory

(Spielberger et al., 1983), which consists of 40 items that combine to a state anxiety score. After completing the questionnaire, participants were seated in a comfortable chair facing a computer monitor at a distance of 0.8 m. They were told that they would hear a sequence of spoken syllables, but that these syllables were unimportant and should be ignored. Participants were asked to attend to a silent, subtitled movie played on the monitor. During the movie, the auditory stimuli were delivered over ear-insert headphones connected to the sound card of the stimulus presentation computer. The sound volume was set at a comfortable level and kept constant for the duration of the experiment.

The EEG was recorded from 64 electrodes mounted in an elastic cap according to the modified 10–20 system. The electro-oculogram (EOG) was recorded from four electrodes attached above and below the right eye and at the outer canthus of each eye. Additionally, one recording electrode was placed on the nose tip. Heart rate was measured by two electrodes attached to the left and right forearm, respectively. The data were recorded at 256 Hz with the ActiveTwo system from Biosemi, which uses a common mode sense active electrode for initial referencing.

2.4. Data analysis

EEG data were processed with EEGLAB (Delorme and Makeig, 2004). The scalp recordings were re-referenced against the nose recording and a 0.5–20 Hz bandpass filter was applied. The continuous data were epoched and baseline corrected using a 200 ms pre-stimulus baseline and an 800 ms time window starting from stimulus onset. Non-typical artifactual epochs caused by drifts or muscle movements were rejected automatically. Infomax, an independent component analysis algorithm implemented in EEGLAB, was applied to the remaining data and components reflecting typical artifacts (i.e., horizontal and vertical eye movements and eye blinks) were removed. Back-projected single trials were screened visually for residual artifacts. To address potential acoustic confounds inherent in a comparison between neutral and angry vocalizations, two difference waves were computed by subtracting neutral standards from neutral deviants and angry standards from angry deviants. Both difference waves revealed an MMN that peaked between 100 and 300 ms following stimulus onset. Accordingly, a 100–300 ms time window was selected to search for the minimum voltage (i.e., MMN) across electrodes, conditions, and participants. The MMN peaked at 210 ms ($SD = 52$) following stimulus onset. A 104 ms time window, centred around the peak of the MMN, was used to obtain individual MMN maxima. These maxima were subjected to an ANOVA with *Condition* (Angry/Neutral), *Region* (Anterior/Posterior), and *Laterality* (Left, Centre, Right) as repeated measures factors. The factors *Region* and *Laterality* comprised the following subgroups of electrodes: anterior-left: AF7 F5 FC5 C5; anterior-middle: AFZ FZ FCZ CZ; anterior-right: AF8 F6 FC6 C6; posterior-left: CP5 P5 PO7 O1; posterior-middle: CPZ PZ POZ OZ; posterior-right: CP6 P6 PO8 O2. This selection of electrodes ensured that the tested subgroups contained equal numbers of electrodes while providing a broad scalp coverage that enabled assessment of topographical differences.

A bipolar HR signal was derived by subtracting the recording from the left forearm from that of the right forearm (Jennings et al., 1981; Bradley and Lang, 2000). After band pass filtering (0.8–40 Hz), heart beats were identified using an R-peak detection algorithm (Mateo and Laguna, 2003) and visually corrected for potentially erroneous R-peak detection. Instantaneous HR was computed from inter-beat intervals and re-sampled at 4 Hz using linear interpolation (Berntson et al., 1997). For each condition, event-related time courses were computed over a 7 s post-stimulus interval and baseline corrected by subtracting the mean signal of two sample points prior to stimulus onset. As for the ERP anal-

ysis, time courses for angry standards were subtracted from those for angry deviants and time courses for neutral standards were subtracted from those for neutral deviants. The resulting difference waves showed a deceleration peak between 0.5 and 1.5 s and an acceleration peak between 4 and 5 s following stimulus onset (see Fig. 1). Mean values in these time windows were used for statistical analysis.

3. Results

3.1. MMN

ERP analysis revealed a *Condition* by *Region* by *Hemisphere* interaction ($F(1, 19) = 3.4, p < .05$). Follow-up analysis indicated that the *Condition* effect varied as a function of Hemisphere over anterior ($F(1, 19) = 4.8, p < .05$) but not posterior regions ($p > .2$). Furthermore, over anterior regions, the *Condition* effect was significant over the right hemisphere ($F(1, 19) = 10.7, p < .01$), approached significance over central electrode sites ($F(1, 19) = 3.1, p = .09$) and was non-significant over the left hemisphere ($p > .2$; Figs. 2–4).

3.2. HR and the MMN

The relationship between HR and the MMN is illustrated in Fig. 4. A two-tailed Pearson correlation analysis for HR deceleration and the MMN was non-significant (all $ps > .2$). However, HR acceleration was significantly larger with increasing MMN amplitudes ($r = -0.51, p < .05$). Although individual analysis for each deviant condition suggested that this effect was more strongly driven by angry ($r = -0.42, p = .06$) as compared to neutral deviants ($r = -0.33, p = .15$), a comparison of independent correlation coefficients was non-significant ($p > .2$; Steiger, 1980). Additionally, a one-way ANOVA comparing mean HR in the deceleration and acceleration time windows suggested that the angry and the neutral deviant condition were comparable ($p > .2$).

3.3. Anxiety and the MMN

Averaged state anxiety scores were 38.05 ($SD = 9.52$) and thus comparable with published norms (Spielberger et al., 1983). A two-tailed Pearson correlation analysis indicated that the MMN amplitude was significantly larger with increasing anxiety ($r = -0.55, p < .05$). Separate analysis of the angry voice condition

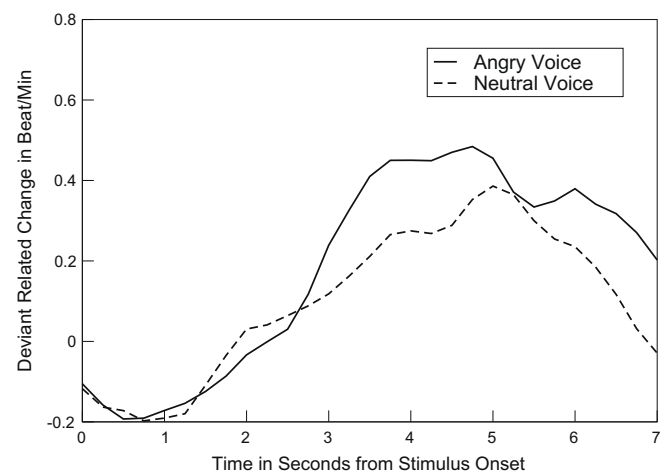


Fig. 1. Subtracting heart rate time courses obtained for standards from those obtained in response to deviants resulted in the difference waves presented here. The solid line represents the grand average of the angry voice condition whereas the dashed line represents the grand average of neutral voice condition.

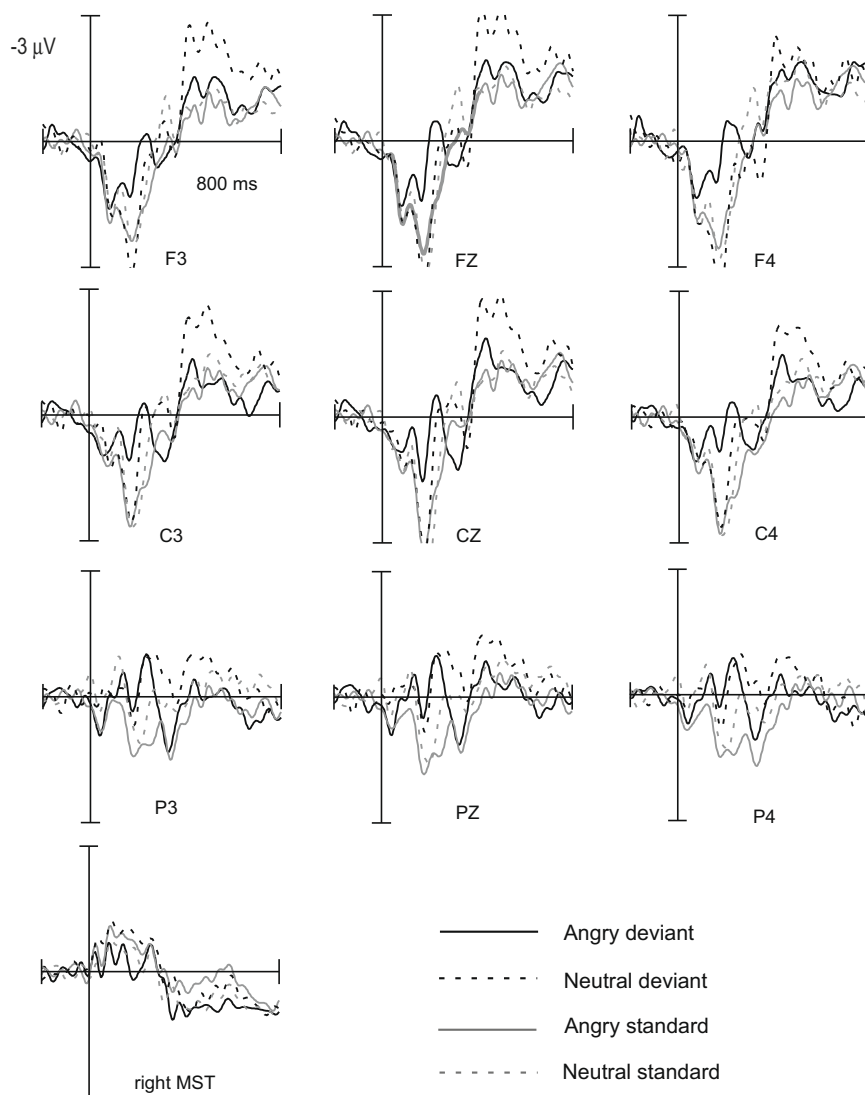


Fig. 2. Grand average ERP traces time-locked to stimulus onset for neutral and angry voice syllables.

revealed that the MMN was larger in individuals with greater state anxiety ($r = -0.62$, $p < .01$). A similar relationship in the neutral voice condition was non-significant ($r = -0.25$, $p = .27$). A comparison of dependent correlation coefficients (Meng et al., 1992) suggested that the angry voice effect and the neutral voice effect were comparable ($p > .2$; Fig. 4).

3.4. Anxiety and heart rate

Finally, we assessed the relationship between state anxiety and the autonomic response to deviants. An analysis using mean HR responses to the deviant conditions was non-significant for HR deceleration ($p > .2$) and approached significance for HR acceleration

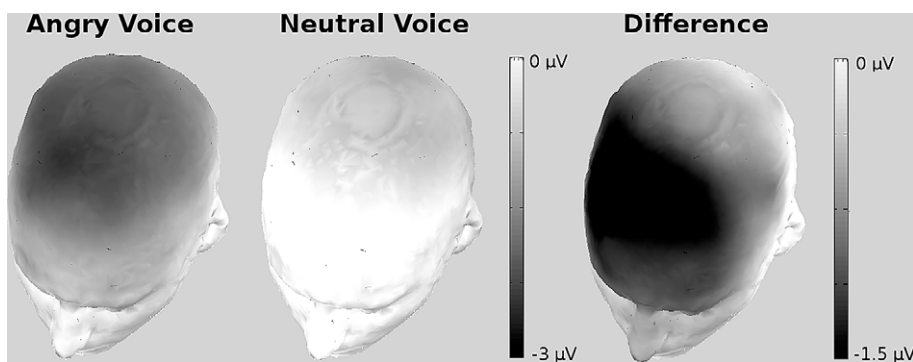


Fig. 3. The figure presents grand average ERP difference maps in microvolt. The first and the second head from the left, illustrate maps derived by subtracting angry standards from angry deviants and neutral standards from neutral deviants, respectively. The difference between both maps is illustrated on the rightmost head. Subtractions were performed on mean MMN amplitudes obtained from a 50 ms centred around the MMN peak.

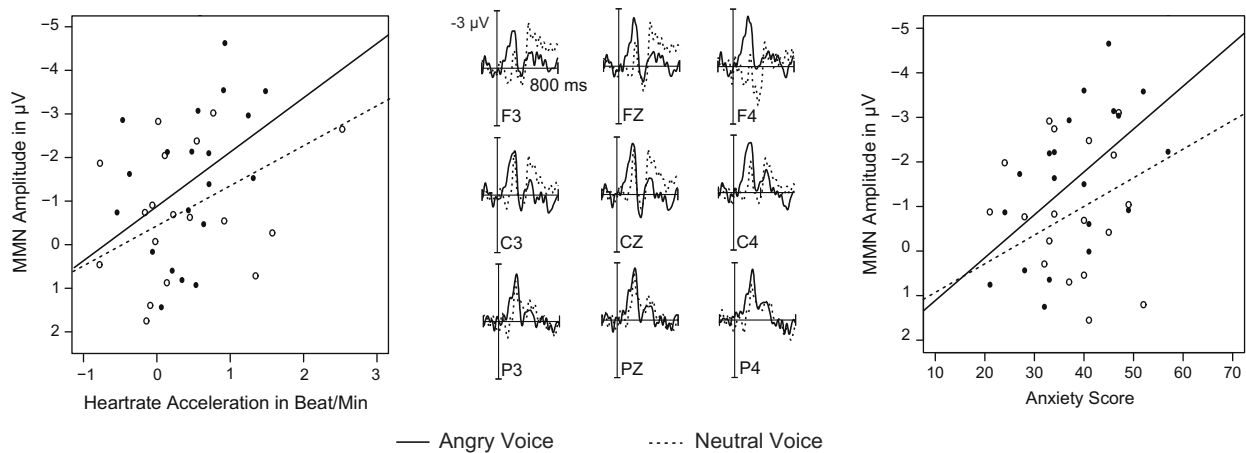


Fig. 4. Grand average ERP difference waves computed by subtracting emotional standards from emotional deviants are represented by a solid line. ERP difference waves computed by subtracting neutral standards from neutral deviants are represented by a dashed line. ERP traces of these difference waves are shown in the middle of the figure. The relationship between mean ERP differences and heart rate is illustrated on the left. The relationship between mean ERP differences and anxiety is illustrated on the right. Open circles represent individual subject values for the neutral condition. Closed circles represent individual subject values for the emotional condition.

($r = 0.42$, $p = .06$). Follow-up analysis for each deviant condition indicated that the latter effect was driven by angry voice deviants ($r = 0.49$, $p < .05$). The relationship between anxiety and HR changes to neutral voice deviants was non-significant ($r = 0.3$, $p = .19$). Again, however, a comparison of dependent correlation coefficients suggested that the angry voice effect did not differ significantly from the neutral voice effect ($p > .2$).

4. Discussion

The present study set out to investigate the relationship between autonomic arousal, anxiety and auditory change detection as reflected by the MMN. Moreover, of particular interest was whether and how this relationship is modified by the emotional significance of auditory change. In line with previous work (Schirmer et al., 2005), we found that the MMN was larger for threat as compared to neutral sound deviants and that this difference was maximal over right frontal leads. Given neuroimaging evidence of the right frontal lobe playing a particular role in avoidance related negative emotions (e.g., Harmon-Jones, 2003; Davidson, 2004), one may speculate that the observed lateralization reflects the elicitation of such an emotion by the angry voice deviants. However, given that a similar lateralization was observed previously in response to happy voice deviants (Schirmer et al., 2005), it may result from an emotion-unspecific frontal activation of attentional or prosodic processes (Schirmer and Kotz, 2006).

The MMN elicited in the present study was significantly correlated with autonomic arousal as assessed through HR changes. In previous work, a characteristic pattern of HR deceleration and acceleration has been observed in response to visual and auditory stimuli and this pattern has been linked to activity of both the central and the autonomic nervous system. Early research associated HR deceleration with attentional orienting towards a provoking event (Lacey and Lacey, 1980; Bradley et al., 2001) as well as combined sympathetic and parasympathetic activation (Bradley et al., 2001). In contrast, HR acceleration was thought to reflect a defensive response to attended stimuli and the mobilization of bodily resources via continued sympathetic innervation (Bradley et al., 2001).

Recent evidence has challenged this view. Researchers reported HR changes for masked stimuli, which participants could not consciously perceive (Katkin et al., 2001). Thus, the elicitation of these changes does not seem to require attentive stimulus processing.

Moreover, if HR deceleration was indeed a correlate for attentional orienting towards the eliciting stimulus, one might expect increased HR deceleration to be associated with improved stimulus memory. However, rather than resulting in memory improvement, researchers found that HR deceleration was associated with later forgetting (Harrison and Turpin, 2003). Based on these and other findings, it has been proposed that the autonomic nervous system receives input from pre-attentive mechanisms and that HR changes can be elicited without conscious awareness of the eliciting event (Harrison and Turpin, 2003).

The present results support this position. They point to a positive correlation between the MMN and HR acceleration suggesting that pre-attentive auditory change detection can induce sympathetic activation. As this correlation showed when collapsing deviant conditions and a comparison between correlation coefficients for the angry and neutral voice conditions was non-significant, one may conclude that complex auditory change in and of itself is important enough to mobilize the peripheral system. Moreover, co-variation between the mechanism driving the MMN and HR seems to exist regardless of a deviant's emotional significance. However, given that individual correlations approached significance for the angry voice deviant and were non-significant for the neutral voice deviant, emotionality may enhance the co-variation between HR and the MMN. This position is in line with the observation that angry deviants are more likely than neutral deviants to activate the amygdala (Schirmer et al., 2008), a structure that is known to influence autonomic arousal (Phelps and LeDoux, 2005). Moreover, this position can explain why previous work using only neutral deviants failed to reveal a relationship between HR and the MMN (Lyytinen et al., 1992).

A second aspect of the present study concerned the role of anxiety in modulating early change detection. Past research has indicated that anxious individuals are more likely than non-anxious controls to attend to threat-relevant information. For example, research using the Posner spatial cuing paradigm implicated anxiety in attention holding (Amir et al., 2003; Fox et al., 2001; Yiend and Mathews, 2001). Anxiety did not facilitate responses to valid threat cues but was found to impair an individual's ability to disengage from invalid ones (Amir et al., 2003; Fox et al., 2001; Yiend and Mathews, 2001). These results support the basic idea of cognitive therapy which assumes that pathological anxiety arises from irrational beliefs and cognitive styles (Beck, 1979). Accordingly, they have been interpreted to reflect an influence of control driven processes on an individual's emotion and behaviour. Findings from

other paradigms have not always supported this view. For example, in a masking study, anxious individuals oriented more readily than healthy controls to unconsciously perceived threat (Mogg and Bradley, 1998). In an attentional blink paradigm, threat-relevant information elicited a reduction in the attentional blink of high relative to low-anxious individuals (Fox et al., 2005). Moreover, past electrophysiological investigations, albeit inconsistently, point towards a relationship between anxiety and pre-attentive change detection (Cornwell et al., 2007; Rosburg et al., 2004). Taken together, the evidence supports the notion that besides modulating control driven processes, anxiety also affects stimulus driven processes (Bishop, 2007).

The present study corroborates this position by providing evidence for a relationship between anxiety and physiological responses to unattended deviants as reflected by the MMN and heart rate. Specifically, anxiety was associated with an increased MMN and increased heart rate to voice deviants suggesting that anxiety enhances pre-attentive change detection and associated autonomic reactions. While individual correlations between anxiety and physiological responses were significant for angry vocalizations they failed to reach significance for neutral vocalizations. Thus, one may conclude that state-dependent changes in anxiety are more likely to affect physiological responses to emotionally significant as compared to neutral change. However, as a comparison of correlation coefficients failed to differentiate between the neutral and angry voice effects it seems that, although weaker, the neutral voice effect resembles the angry voice effect. Thus, as for the relationship between the MMN and HR, the relationship between anxiety and pre-attentive change detection may be stronger and hence more easily observed for change that is emotionally significant.

Overall, the present study replicates and extends previous work on the relationship between the MMN and emotional processing. When contrasting vocal deviants and standards, we observed larger MMN amplitudes for angry as compared to neutral tone of voice. This is in line with evidence indicating that the MMN is elicited not only to simple physical change, but also to changes in the integrated percept arising from auditory input (Gomes et al., 1997). Moreover, our results confirm the idea that the significance of this percept for the individual can modulate MMN amplitude (Goydke et al., 2004; De Baene et al., 2004; Schirmer et al., 2005, 2007). However, whether significance for the individual modulates MMN amplitude seems to depend on how significance was acquired. Specifically, if the personal significance of an auditory object was learned and is thus reliant on long-term memory representations, not the MMN but a process following the MMN appears to be enhanced (Royer et al., 2007). For example, compared to other ring-tones, the ringing of one's own cell phone does not increase MMN amplitude, but increases a subsequent posterior negativity (Royer et al., 2007). In contrast, if the personal significance of an auditory object is intrinsic and does not require learning, as in the case of vocal threat (Bachorowski and Owren, 2003), the MMN amplitude itself may be enhanced. In line with this, we show that vocal expressions of threat, which induce negative emotions due to their unpleasant physical characteristics (Bachorowski and Owren, 2003), elicit a greater MMN than neutral vocalizations. The MMN response to threat is further enhanced if an individual feels anxious thus indicating that state anxiety raises an individual's sensitivity to changes in the environment. Together with the fact that ensuing autonomic arousal correlates with both anxiety and the MMN, these results strengthen the role of the MMN as a marker for emotional processing. Moreover, they support the proposition that pre-attentive change detection as reflected by the MMN represents a mechanism by which unattended threat in the environment can excite cognitive and emotional responses that prepare the body for action.

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